

LAST NAME: _____

Practice Quiz 04B — MATH 2055 — Fall 2026

Practice for Quiz 04

The following is closed book and closed note. Don't assume the grader will know what you mean. Use full sentences and provide complete and correct solutions.

Revised using GPT-6 Astra in the Codex harness.

1. (a) Disprove the identity

$$(A \cup B) \setminus C = A \cup (B \setminus C)$$

by giving finite sets A, B, C and computing both sides.

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- (b) Express $(A \cup B) \setminus C$ correctly as a union of two set differences. Prove your identity by a chain of membership equivalences, naming the definitions and logical law used.